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GEO-Insight

Which humanitarian crises are most overlooked?

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Repository: github.com/BenBullinger/GEO-Insight

THE QUESTION

Relative to documented need, which crises are *most overlooked*?

Humanitarian coordinators and donor advisors need a defensible answer every cycle. No existing tool gives them one.

The scale of the gap

$33B

2026 GHO appeal
to reach 135M people

$12B

Raised in 2025
the lowest in a decade

19

Severely underfunded contexts
up from 8 in 2021

Sudan fell from 69 % covered (2024) to 35 % (2025). Yemen, Somalia, and DRC sit near 22–24 %.

THE ANALYTICAL PROBLEM

When *every* response is underfunded, "underfunded" stops being useful.

A single coverage ratio can't discriminate between 19 contexts that are all below 50 %. The interesting variation is no longer *whether* a crisis is underfunded — but *how*, and *for whom*.

PART I

A sharper question.

Three ways to be overlooked

Existing tools collapse these into one number. They aren't the same crisis.

HIGH NEED · LOW FUNDING · LOW ATTENTION

Forgotten

Angola drought, 2023 — 7M affected. Just 1,049 news articles worldwide. (*Barbie*, same year: 273,279.)

HIGH NEED · LOW FUNDING · HIGH ATTENTION

Contested

Sudan, 2025 — world's largest displacement crisis, on every front page. Coverage still dropped from 69 % to 35 %.

ADEQUATE AGGREGATE · STARVED CLUSTER

Sector-starved

A crisis at 60 % aggregate coverage can hide a **health cluster at 20 %** while food is overfunded. The total masks the starved part.

From one number to a joint distribution

EXISTING TOOLS

0.37

A scalar coverage ratio.
Cannot distinguish the three kinds.

→

GEO-INSIGHT

intra-crisis equity
× inter-stakeholder disagreement

over a six-signal score

A distribution that distinguishes *forgotten*, *contested*, and *sector-starved*.

Each new axis is anchored in an explicit gap in the 2015–2026 humanitarian-analytics literature (Abdulrashid 2026 §7.3; Sahebjamnia 2017 §6; Vargas Florez 2015 Eqs. 9–12).

How the score works

For each crisis, **six signals** are weighted by a **donor profile** and aggregated.

$$G_p(u) \;=\; \sum_{i=1}^{6} w_{p,i} \cdot U_i(\tilde{a}_i(u))$$

$G_p(u)$ gap score for crisis u under profile p

$w_{p,i}$ profile p's weight on signal i

$U_i(\cdot)$ utility transform for signal i

$\tilde{a}_i(u)$ normalised signal value for u

Six signals: coverage shortfall · per-PIN funding gap · need intensity · severity · donor concentration · *intra-crisis equity Gini (new)*.
Four profiles: CERF-, ECHO-, USAID-, NGO-flavoured.

Four kinds of overlooked, decoded

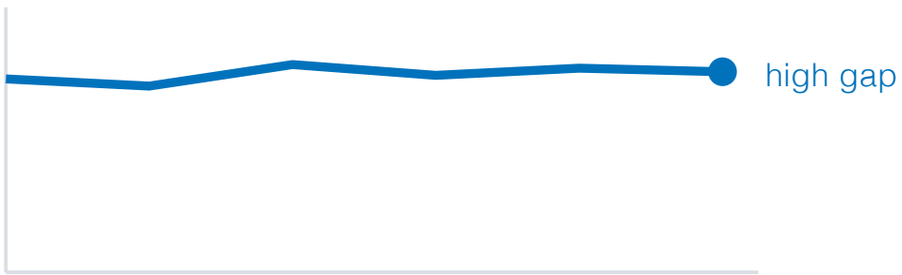
Rank each crisis under *every* donor profile. Combine the spread of ranks with the sector inequality.

	LOW DISAGREEMENT	HIGH DISAGREEMENT
HIGH EQUITY GAP	**Consensus-starved-sector** Clearest operational target. One cluster missing; every mandate agrees.	**Contested *and* sector-starved** Highest advocacy priority. Politically contested AND a missing cluster.
LOW EQUITY GAP	**Consensus-overlooked** CERF UFE candidates. Everyone agrees — underfunded across the board.	**Contested but balanced** Reveals donor politics. Priorities diverge; no missing cluster.

A third axis: time

A crisis overlooked for five years needs different policy than one that just tipped. We separate the two.

Chronic



Persistent underfunding across years. **Response:** multi-year pooled funding.

Acute

Recent collapse from baseline. **Response:** CERF rapid-response window.

Chronic = k-year rolling mean of the gap. Acute = deviation of this year from that rolling mean.

PART II

Does it agree with reality?

We don't validate against ourselves

Two lists, human-curated by independent organisations, without using our method:

OCHA

CERF Underfunded Emergencies

Twice-yearly allocation to "*protracted or neglected humanitarian crises where funding is low but vulnerability is high.*"

→ should match our *consensus-overlooked* quadrant.

CARE

Breaking the Silence

Annual list of the 10 most *under-reported* humanitarian crises. Measured by media attention — not funding.

→ should match our *sector-starved / forgotten* axis.

Metrics: Overlap@10, Spearman ρ, sensitivity to profile choice. One benchmark per axis.

What comes out — top-ranked by typology

[Populated from implementation output]

#	Country	Median rank	IQR	Gini	Typology
1	[—]	[n/N]	[Δ]	[0.xx]	consensus-overlooked
2	[—]	[n/N]	[Δ]	[0.xx]	contested-starved
3	[—]	[n/N]	[Δ]	[0.xx]	consensus-starved-sector

What this is *not*

NOT AUTOMATION

The tool ranks. Humans decide. A score is a hypothesis, never a verdict.

NOT A SINGLE NUMBER

Four profiles, four ranks. We always report the median *and* the IQR.

NOT IMPUTED

Stale HNO, missing clusters, plan-less crises are flagged — never silently filled.

NOT NEUTRAL

We *surface* donor disagreement as signal, instead of hiding it behind consensus.

NOT COMPLETE

~20 countries are full-stack. The rest degrade gracefully; we say so.

NOT PRECISE

Two significant figures. A score of 0.84 is not meaningfully distinct from 0.82.

THANK YOU

Questions?

`proposal/proposal.pdf` · `github.com/BenBullinger/GEO-Insight`